Playing with Constraints: Designing Inclusive Games and Controllers

Priscilla Y. Lo
School of Computer Science
University of Bristol
Bristol, United Kingdom
priscilla.lo@bristol.ac.uk

Grace Stangroome
School of Computer Science
University of Bristol
Bristol, United Kingdom
g.stangroome.2019@bristol.ac.uk

Vishal Joshi
School of Computer Science
University of Bristol
Bristol, United Kingdom
we19383@bristol.ac.uk

Jack Burnett
School of Computer Science
University of Bristol
Bristol, United Kingdom
jack.burnett@bristol.ac.uk

David Thue
School of Information Technology
Carleton University
Ottawa, Ontario, Canada
david.thue@carleton.ca

Oussama Metatla
School of Computer Science
University of Bristol
Bristol, United Kingdom
o.metatla@bristol.ac.uk



Figure 1: Images from a game jam (left to right): playing, prototyping, and discussing.

Abstract

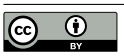
How can we design inclusive games that appeal to gamers with different abilities? This workshop invites the CHI PLAY community to play with constraints and conceptualize games that enable forms of play beyond conventional patterns. We build on our prior work that applies the concept of "impaired technologies" as a design provocation in a game jam to encourage alternative perspectives on play. Participants will create multiplayer games that are played with impaired technologies or, as an extension, that use novel interactions. We aim to engage researchers and practitioners interested in designing playful prototypes, including digital games, immersive experiences, tangible toys, and game controllers.

CCS Concepts

• **Human-centered computing** → *Interaction design*; *Interaction paradigms*; *Accessibility*.

Keywords

game jam, inclusion, accessibility, playful embodiment



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1 Introduction and Motivation

Video games often pose barriers for disabled gamers [8], with accessibility often being treated as an afterthought or an additional feature. When companies do not prioritize disabled access, individuals take on additional and tiring access labour to craft fun experiences, which often increases frustration, instead, leaving disabled gamers locked out of playing games that appeal to them [3, 6]. Access labour can look like rebinding keys or exerting more effort to turn a doorknob [2, p. 50]. Although games specifically designed for people with disabilities (e.g., audio-only games) are more accessible to everyone [5], they may be perceived as less engaging or more challenging for able-bodied people. We propose this workshop to inspire the CHI PLAY community to design inclusive games playable by people with mixed-abilities (disabled and non-disabled), while considering alternative and inclusive ways of playing together.

When gamers encounter gameplay challenges, they may either give up or adapt their approach. Adaptation takes many forms, from the appropriation of tools to rethinking strategy, including cheating; these adaptations can inform new forms of play where

such behaviour becomes a deliberate mechanic. Activist affordances [2] and glitch feminism [9] both suggest ideas that, in the face of resistance and constraints, humans develop new ideas and directions (e.g., through iterative adaptation and strategic experimentation [4]). Within this framing and extending our prior work [7], we present a provocation in which gaming peripherals are impaired (e.g., cracked screens, missing buttons on controllers, and unstable audio). Participants will design multiplayer games for these devices, with an opportunity to additionally prototype novel interactions for those games. By embracing constraints and obstacles, we aim to channel creativity towards innovations that extend beyond conventional boundaries of games that rely on audiovisual media.

2 Objectives

We aim to explore how designers of play can gain new perspectives by adapting their strategies and behaviours, using play as an approach to support public understanding of accessibility and as a source of inspiration for inclusive design. This workshop aims to motivate participants to consider new perspectives, encourage diverse future technologies, and build community via a Discord server, all through the design of inclusive games. We aim to submit an article on the workshop outcomes to ACM Interactions, and the games will be shared on our workshop website gallery.

3 Workshop Format

We expect the workshop to last for 4 hours. This is to allow time to experience gameplay with impaired technologies to prime participants' design thinking, and facilitate a group discussion to increase the group's understanding and reflections about the concept. As outcomes, participants will conceptualize multiplayer games, which can be extended with novel interactions (e.g., custom controllers for the games). Admission to the workshop will be primarily based on pre-registration, but walk-ins will also be welcome if space permits. We can accommodate up to six groups of five people within the planned time (up to 30 participants in total).

To join, we will ask participants to complete a short web form to help us understand their backgrounds (designer/developer/UX), interests (digital/physical prototyping), access needs, and preferred attendance method (online/in-person). This creates an accessible barrier to entry, and people who complete the form in advance will be guaranteed a space. If there are more applicants than space allows, we will balance the number of designers with developers; those welcome to attend will be emailed in advance.

To facilitate online participation, we will use a Discord server to build community and simplify text and image interaction between in-person and online participants. All participants can add Discord roles to match their self-identified backgrounds and interests. A Microsoft Teams video call with live captioning will remain open throughout the workshop, and we will create virtual rooms for online group interaction.

3.1 Accessibility

At the beginning of the session, participants will be reminded of several important guidelines for a smooth workshop experience: to take as many breaks as they need, to introduce themselves before speaking, and to create a safe space by not calling on specific individuals to contribute. Live captioning will also be available. We will plan for alternative forms of involvement, such as writing on post-it notes, using the online shared collaborative space Padlet [1] (a visual collaboration software), and working in small teams or individually. The workshop is available in hybrid format. Additionally, we invite participants to bring and use pillows, blankets, noise-cancelling headphones, or other bodily extensions that make them more comfortable.

3.2 Activities and Interactive Components

We will provide the games, impaired technologies, and craft materials required for participation. Online participants can create their own constraints by using post-it notes or paper to cover parts of their monitor. We will provide examples and ask online participants to take a photo of their post-it note arrangement and share it. Participants will need to bring their own laptops if they wish to prototype their games digitally.

The times in the following subheadings total less than 4 hours, allowing a 10-minute buffer between sections and ensuring all attendees and presenters can take breaks. Upon arrival, participants will be welcome to attach coloured/shaped stickers to their name badge to self-identify their various backgrounds (e.g., designer, developer) and interests (e.g., digital, physical). These stickers and Discord roles will help us create teams for Part 2.

3.2.1 Part 1: Gaming with Impaired Technologies (45 Minutes). Participants will be invited to play games with various impaired technologies provided by the organizers. Participants can also spectate other games in play by other participants, which will be live-streamed via Teams for online participants. Throughout this gaming session, the organizers and participants will use Padlet [1] or post-it notes to write down their impressions about the experience of playing using impaired technologies. Some potential discussion questions include:

- In what way(s) did you feel constrained playing the game?
- What did you feel you were able to do in the game?
- How did you work with the constraints to make new possible actions and fun?

Finally, we will propose teams for participants who completed the web form; we will ask that all participants form teams or confirm individual involvement at the end of this phase. Participants will subsequently collaborate within these teams to create new games.

3.2.2 Part 2: Game Conceptualization or Physical Prototyping (75 Minutes). Participant teams will be asked to conceptualize their own **multiplayer** games that are played with **impaired technologies**, which may be extended using **novel interactions** (e.g., unconventional ways of interacting with games, custom controllers, etc.). They can create and communicate their game concepts by vibe-coding, prototyping using craft materials (e.g., see Figure 1), or role-playing. We will encourage participants to consider designing beyond what existing controllers look like, and consider devices that can both accept inputs and return outputs.

3.2.3 Part 3: Pitch and Discussion (60 Minutes). Each team will have a few minutes to demonstrate their designs from Part 2 to all workshop participants. The goal is to present their inclusive games

as a pitch that appeals to mixed-ability gamers (disabled and non-disabled). They will be invited to share their designs with either presentations or bodystorming simulations. After each demonstration, all workshop participants will then discuss the following:

- How were constraints incorporated into gameplay? What unexpected interactions emerged?
- Who (if anyone) is excluded from playing this game? What adaptations could be made to support them?
- What would be necessary for this game to be realized? How can commercial games learn from these ideas?

Throughout the workshop, we will continuously integrate input from the online, anonymous written space on Padlet [1] or Post-it notes into the spoken discussion.

3.2.4 Part 4: Conclusion (30 Minutes). In closing, we will offer participants the opportunity to share any remaining thoughts and compile perspectives from the alternative forms of participation. These final insights will also be documented on Padlet [1]. With participants' permission, we will upload their game concepts to a gallery on the workshop website, for sharing and future reference.

4 Organizers

Priscilla Y. Lo (she/her) is a PhD student at the Diverse-Ability Interaction Lab within the Bristol Interaction Group (BIG) at the University of Bristol. She has a background in games design and and her prior research ranges across games, AI, and cross-sensory correspondences. Her current research explores leveraging playful inclusive public engagement events (e.g., game jams) to encourage people with different abilities to imagine alternative sensory interactions beyond traditional audiovisual media.

Grace Stangroome (she/her) is a PhD student at BIG, University of Bristol. Her research aims to understand how HCI can be used to improve co-working in mixed-ability (disabled and non-disabled) university student groups.

Vishal Joshi (he/him) is a PhD student in the Interactive AI Centre for Doctoral Training at the University of Bristol, affiliated with BIG. His research aims to understand cooperation between LLM-agents as well as human-LLM cooperation, with a particular focus in table-top roleplaying game settings.

Jack Burnett (he/him) is a PhD student in the Interactive AI Centre for Doctoral Training at the University of Bristol. His research focuses on democratizing game hardware to enable low-tech approaches to individualized solutions, including accessible and adaptive controllers.

David Thue (he/him) is an Associate Professor of Information Technology at Carleton University, where he leads the RISE (Reasoning for Interactive Systems and Experiences) research group. His work spans games and other interactive technologies, with a focus on the intersection between Artificial Intelligence and Human-Computer Interaction.

Oussama Metatla (he/him) is an Associate Professor of Human-Computer Interaction at the University of Bristol, where he co-directs BIG and leads the Diverse-Ability Interaction Lab. His research explores cross-sensory interaction, cross-modal perception, and embodied cognition to co-design technologies that enable more inclusive interactions between disabled and non-disabled people.

References

[1] 2026. Padlet - Visual Collaboration for Creative Work and Education. https://padlet.com/

[2] Arseli Dokumaci. 2023. *Activist affordances: How disabled people improvise more habitable worlds.* Duke University Press.

[3] Jay T. Dolmage. 2017. *Academic Ableism: Disability and Higher Education.* University of Michigan Press, Ann Arbor, Michigan, United States. doi:10.3998/mpub.9708722

[4] Tibor Guzsvinecz and Judit Szűcs. 2024. Game Theory and Strategic Decision-Making in the Dark Souls Games. In *Game Theory - Computational Aspects and Applications.* IntechOpen. doi:10.5772/intechopen.1008456

[5] Carmel Heydarian. 2020. *The Curb-Cut Effect and its Interplay with Video Games.* A Research Report for the Degree Master of Science in Graphic Information Technology. Arizona State University.

[6] Erin Higgins, Zaria Oliver, and Foad Hamidi. 2023. Towards a Social Justice Aligned Makerspace: Co-designing Custom Assistive Technology within a University Ecosystem. In *The 25th International ACM SIGACCESS Conference on Computers and Accessibility.* ACM, New York NY USA, 1–13. doi:10.1145/3597638.3608393

[7] Priscilla Y. Lo, Grace Stangroome, Vishal Joshi, Jack Burnett, David Thue, Dan Bennett, and Oussama Metatla. 2026. Designing with Impaired Technologies: A Game Jam of Constraints and Affordance Shrinkage. In *Companion Proceedings of the Annual Symposium on Computer-Human Interaction in Play (CHI PLAY Companion '26), November 02–05, 2026, York, United Kingdom* (York, United Kingdom) *(CHI PLAY '26).* Association for Computing Machinery, New York, NY, USA, 8 pages. doi:10.1145/3800965.3834287

[8] Jesse J Martinez, Jon E. Froehlich, and James Fogarty. 2024. Playing on Hard Mode: Accessibility, Difficulty and Joy in Video Game Adoption for Gamers with Disabilities. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24).* Association for Computing Machinery, New York, NY, USA, Article 524, 17 pages. doi:10.1145/3613904.3642804

[9] Legacy Russell. 2020. *Glitch Feminism.* Verso, New York, NY, USA. https://www.versobooks.com/en-gb/products/460-glitch-feminism